

Lab: The Social Brain -- Mirror Neurons, Empathy, and Biological Communication

Objective

Investigate how the brain implements **social communication** through mirror neurons, empathy circuits, and hormonal signaling, connecting biological mechanisms to the Active Inference concept of model synchronization between agents.

Prerequisites

- Completed Biology modules on Systems through Learning
- Read the module on Mirror Neurons, Empathy, and Social Brains

Part 1: Mirror Neuron Investigation

Goal: Understand how mirror neurons enable social prediction.

Research mirror neurons and answer:

1. What are mirror neurons? Where were they first discovered?
2. What do mirror neurons do when you watch someone else perform an action?
3. How might mirror neurons help you predict what another person is about to do?
4. How does this relate to Active Inference? (Hint: your brain runs its own motor model when observing someone else, generating predictions about their actions.)

Write your response here

Part 2: Empathy Observation Exercise

Goal: Observe biological empathy responses in yourself.

Watch a video of someone experiencing a strong emotion (joy, pain, fear, sadness). While watching, pay attention to your own body:

What You Observe in the Video	What You Feel in Your Own Body	Biological Mechanism
Someone wincing in pain		
Someone laughing		
Someone crying		
Someone expressing fear		

Why does your body mirror the emotions of others? How is this biological empathy related to the Active Inference idea of modeling another agent's internal states?

Write your response here

Part 3: Chemical Communication

Goal: Map the hormones and neurotransmitters involved in social bonding.

Chemical	Where It Is Produced	What It Does Socially	Active Inference Role
Oxytocin			
Cortisol			
Dopamine			
Endorphins			

How do these chemicals help agents synchronize their internal states?

Write your response here

Part 4: Communication Across Biological Scales

Goal: Compare communication at different biological levels.

Level	Sender	Signal Type	Receiver	What Is Synchronized?
Cell-to-cell	Neuron		Neuron	
Organ system	Endocrine gland		Target tissue	
Organism-to-organism	Human		Human	
Microbiome-to-host	Gut bacteria		Brain (via vagus nerve)	

What do all these communication systems have in common when viewed through Active Inference?

Write your response here

Summary Table

Concept	Biological Mechanism	Active Inference Connection
Mirror neurons	Motor cortex activation during observation	Simulating another agent's model
Empathy	Shared neural circuits for self and other	Predicting another agent's internal states
Hormonal signaling	Oxytocin, cortisol, dopamine	Chemical precision signals for social bonding
Multi-scale communication	Neurons, hormones, behavior, microbiome	Model synchronization at every level